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# Fast Tree Diffusions for Probabilistic Tabular Regression via Flow Matching

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## Abstract

1 We extend Treeffuser, a score-based diffusion model for probabilistic tabular  
2 regression with gradient-boosted trees, in two directions. A score-side recipe—  
3 conditional-mean residualization, EDM-style preconditioning, log-sigma time sam-  
4 pling, and noise-level features—improves aggregate calibration after per-dataset  
5 tuning, reducing 95% coverage error from 0.023 to 0.015 while cutting sampling  
6 time  $6.5\times$  over the published baseline. A flow-matching variant with Gaussian  
7 paths and algebraic score recovery gives the best normalized CRPS among Treef-  
8 fuser rows with a VP path and deterministic five-step Heun ODE, and is another  
9  $2.4\times$  faster than the improved score model, but gives back some coverage. The  
10 common mechanism is better conditioning of the supervised regression problem  
11 seen by the tree ensemble. Under this ten-dataset tuning protocol, the Treeffuser  
12 variants give the best aggregate normalized CRPS and coverage metrics among  
13 displayed rows, though gains over the published baseline are directional rather than  
14 conventionally significant ( $p \approx 0.07$ ), and tuned non-diffusion baselines still win  
15 several datasets on raw CRPS. Stochastic ( $\varepsilon > 0$ ) samplers do not Pareto-dominate  
16 the deterministic corner, so the empirical recommendation is  $\varepsilon = 0$  when aggregate  
17 CRPS and latency are the target, while stochasticity remains a conditional-tail  
18 calibration knob.

## 19 1 Introduction

20 Probabilistic regression on tabular data must combine the predictive strength of gradient-boosted  
21 trees with full predictive distributions usable for calibration, decision-making, and uncertainty  
22 quantification. Treeffuser [Beltrán-Velez et al., 2024] achieves this by training LightGBM [Ke et al.,  
23 2017] regressors to estimate the score of a noise-perturbed conditional density and integrating the  
24 corresponding reverse-time stochastic differential equation (SDE) [Song et al., 2021] at inference. The  
25 framework inherits the calibration properties of score-based diffusion while avoiding neural-network  
26 density estimators.

27 Our notion of tractability is computational rather than likelihood-exact: predictive quantities are  
28 Monte Carlo estimates, but training reduces to supervised tree regression, the final FM row samples  
29 with a five-step Heun ODE, and Eq. (2) recovers scores when stochastic sampling is desired.

30 We develop Treeffuser along two axes: score-side modifications matched to histogram-bin tree  
31 learners, and Gaussian-path flow matching [Lipman et al., 2023, Liu et al., 2023], which removes the  
32 indirect score-to-drift mapping and opens access to stochastic interpolants [Albergo et al., 2023] with  
33 non-negative stochasticity schedule  $\varepsilon(t)$ .

34 **Contributions.** We make three contributions:

1. *Score-side modifications for tree-based diffusion*: conditional-mean residualization, EDM preconditioning, log- $\sigma$  time sampling, and an explicit log- $\sigma$  feature for LightGBM score regression (§2.1).
2. *Flow matching for Treeffuser*: a Gaussian-path FM trainer with score recovery when stochastic-interpolant sampling or endpoint-error analysis requires it (§2.2, Lemma 1).
3. *Regression-conditioning unification*: regression conditioning for the tree ensemble as the bottleneck behind both score- and flow-side gains (§4, Appendix A).

Across ten UCI benchmarks, with one fold for tuning and five for evaluation, the score-side recipe gives the best aggregate calibration and CRPS skill score among Treeffuser rows, while FM gives the best aggregate normalized CRPS and fastest Treeffuser sampler. The shared lesson is that noising path, parameterization, training distribution, features, and sampler must make the induced supervised problem well matched to histogram splits.

## 2 Method

### 2.1 Score-Side Modifications

The published Treeffuser uses VESDE marginals with uniform  $t$  sampling, a noise-prediction target, and LightGBM features  $[y_t, X, t]$ . We change four pieces: response/input conditioning, allocation of training rows over noise levels, explicit noise-level features, and sampler efficiency.

**Conditional-mean residualization.** We fit a cross-validated LightGBM mean predictor  $\hat{\mu}(x)$  and apply diffusion to  $y - \hat{\mu}(x)$  rather than  $y$ , moving conditional location out of the score-regression target. The residualizer uses a high-capacity configuration selected from a prior diagnostic sweep (Appendix D) and then fixed; it is not tuned jointly with the diffusion or flow hyperparameters. On large datasets this fixed residualizer can become a bottleneck once the inner model has enough data to represent the conditional mean directly.

**EDM preconditioning.** Following Karras et al. [2022], we predict a denoised target  $D_\theta(y_t, t)$  scaled to unit variance across noise levels and reconstruct the score  $s(y_t, t) = (D_\theta - y_t)/\sigma(t)^2$ . This stabilizes the regression target for trees, which would otherwise need to extrapolate noise-prediction magnitudes across  $t$ .

**Log-sigma time sampling.** We draw  $\log \sigma(t) \sim \mathcal{N}(-1.2, 1.2^2)$ , clip to the SDE’s achievable range, and invert. Because trees bin features by data density, the training-time distribution directly controls where the score model spends capacity.

**Log-sigma noise feature.** The feature vector for LightGBM becomes  $[y_t, X, t, \log \sigma(t)]$ , giving the regressor an explicit noise-level signal aligned with EDM scaling.

### 2.2 Flow Matching with Gaussian Paths

A Gaussian flow path interpolates data  $y_0$  and a standard-normal prior  $z$  via

$$y_t = \alpha(t) y_0 + \beta(t) z, \quad t \in [0, 1], \quad (1)$$

with boundary conditions  $\alpha(0) = 1, \alpha(1) = 0, \beta(0) = 0, \beta(1) = 1$ . We compare three paths: *linear* ( $\alpha = 1 - t, \beta = t$ ), *trigonometric* ( $\alpha = \cos(\pi t/2), \beta = \sin(\pi t/2)$ ), and a *VP* schedule with linear  $\beta_t$  giving  $\alpha^2(t) = \exp(-T(t))$ ,  $T(t) = \frac{1}{2}\beta_{\min}t + \frac{1}{4}(\beta_{\max} - \beta_{\min})t^2$ . The training target is the conditional velocity  $u_t = \alpha'(t)y_0 + \beta'(t)z$ , regressed against the model’s  $v_\theta(y_t, x, t)$ .

**Velocity-to-score formula.** For any Gaussian path with Wronskian  $W(t) = \alpha(t)\beta'(t) - \alpha'(t)\beta(t)$ , the conditional score is algebraically recoverable from the population velocity:

$$\nabla \log p_t(y_t | x) = \frac{\alpha'(t) y_t - \alpha(t) v_\theta(y_t, x, t)}{W(t) \beta(t)}. \quad (2)$$

For linear FM this reduces to  $-(y_t + (1 - t)v)/t$ ; for trig to  $-y_t - (2/\pi) \cot(\pi t/2)v$ . We use this identity operationally: it lets a velocity-trained Treeffuser enter the same stochastic-interpolant sampler family as score matching and exposes the endpoint amplification studied in Appendix A.

78 **Stochastic-interpolant family.** Albergo et al. [2023] show that any non-negative  $\varepsilon(t)$  defines a  
 79 marginal-preserving reverse-time sampler at the population velocity/score:

$$dy_\tau = \left(-v_\theta + \frac{\varepsilon^2}{2} s_\theta\right) d\tau + \varepsilon dW_\tau, \quad (3)$$

80 where  $\tau = 1 - t$  is reverse time and  $s_\theta$  is the score from (2). The choice  $\varepsilon \equiv 0$  recovers the  
 81 deterministic probability-flow ODE; positive  $\varepsilon(t)$  broadens samples, with exact marginal preservation  
 82 only for the true velocity/score. We use the schedule  $\varepsilon(t) = ct$ , which vanishes at the data endpoint  
 83 where the learned score’s  $1/\beta(t)$  factor would otherwise dominate (see Proposition 3).

84 **Why these choices suit trees.** Eq. (2) makes FM compatible with stochastic-interpolant samplers;  
 85 variance-preserving paths keep the tree input  $y_t$  near unit scale, unlike a linear path with  $\text{Var}(y_t) \approx$   
 86  $(1 - t)^2 + t^2$ ; and  $\varepsilon = 0$  avoids the endpoint-sensitive  $1/\beta(t)$  score-error factor (Appendix A).

## 87 2.3 Implementation Notes

88 The FM code reuses Treeffuser’s per-output-dimension LightGBM fitter and feature pipeline; only  
 89 the target and prior change. Sampling uses the same Heun integrator and residualizer as the score  
 90 path, and both training and sampling avoid the singular endpoint with  $t \geq 10^{-5}$ .

## 91 3 Experiments

92 **Datasets and protocol.** Ten UCI benchmarks (Table 2) are evaluated with a nested tuning protocol.  
 93 We split each dataset into six folds: fold 0 is used only for hyperparameter selection, and folds 1–5  
 94 are evaluation folds. For each dataset/model family, Optuna tunes on the fold-0 train/validation split;  
 95 the selected configuration is then refit and evaluated on each held-out evaluation fold. Evaluation-fold  
 96 training rows range from 256 on yacht to 38,108 on protein. For sample-based methods, each test  
 97 point uses 200 samples; Treeffuser sampler choices are detailed in Appendix B. Metrics are averaged  
 98 over evaluation folds and, for headline summaries, over datasets.

99 **Metrics.** We report CRPS skill score  $\text{CRPSS} = 1 - \text{CRPS}_{\text{model}}/\text{CRPS}_{\text{climatology}}$ , mean rel-CRPS,  
 100 DSS [Dawid, 1984], PIT-KS pass rate at  $\alpha = 0.05$ , absolute coverage error at 50/90/95% nominal  
 101 levels [Gneiting and Raftery, 2007], and sample-generation time. CRPSS and rel-CRPS are the  
 102 aggregate comparison metrics because raw CRPS is target-scale dependent; mean climatological  
 103 CRPS ranges from 0.0043 on naval to 44.8 on diabetes. Timings are measured on one Apple-  
 104 arm64 workstation, exclude fitting, and include generating 200 samples per test point plus inverse  
 105 transformations. They use practical benchmark settings rather than a single-thread cap: Treeffuser  
 106 samples in batches with `n_parallel=20` and LightGBM `n_jobs=-1`; other libraries use their  
 107 configured parallel defaults.

108 **Variants compared.** Three Treeffuser rows compare bundled recipes: (1) PUBLISHED—the original  
 109 baseline with noise-prediction, uniform  $t$ , no residualization; (2) SCORE+—the baseline plus §2.1;  
 110 (3) FM—VP flow matching with residualization and a 5-step Heun ODE. Each row is tuned per  
 111 dataset under the same fold-0 protocol. The residualizer-C configuration used in the headline rows  
 112 was selected from the prior diagnostic sweep in Appendix D and then fixed; it is not tuned jointly with  
 113 the diffusion or flow hyperparameters. We also compare against six per-dataset tuned probabilistic  
 114 baselines: NGBoost [Duan et al., 2020], quantile-regression LightGBM [Ke et al., 2017], CatBoost  
 115 RMSE-with-uncertainty [Prokhorenkova et al., 2018], iBUG [Brophy and Lowd, 2022], a Gaussian  
 116 deep ensemble [Lakshminarayanan et al., 2017], and CARD-style neural diffusion [Han et al., 2022].

### 117 3.1 Headline Result

118 Table 1 reports cross-dataset means under the tuning/evaluation protocol. The score-side recipe lifts  
 119 CRPSS from 0.670 to 0.681, raises PIT-uniformity pass rate from 0.40 to 0.62, reduces 90/95%  
 120 coverage error from 0.044/0.023 to 0.020/0.015, and cuts Treeffuser sample time  $6.5\times$ . FM is  
 121 close on CRPSS, has the best aggregate normalized CRPS among displayed rows, and is another  
 122  $2.4\times$  faster than score<sup>+</sup>, but gives back part of score<sup>+</sup>’s coverage advantage. Per-dataset raw-CRPS  
 123 winners span five families; a Treeffuser variant wins on 6/10 datasets, with the remaining wins split

Table 1: Headline real-data results, mean over 10 UCI benchmarks with fold 0 used for tuning and folds 1–5 used for evaluation. rel-CRPS is the per-dataset CRPS divided by the best displayed variant on that dataset.  $|cE|@q$  is absolute coverage error at the  $q\%$  interval level. The CARD row is a compact benchmark adapter, not a fully optimized CARD pipeline (see Appendix H).

| Variant                              | CRPSS $\uparrow$ | rel-CRPS $\downarrow$ | KS $p > .05 \uparrow$ | $ cE @50 \downarrow$ | $ cE @90 \downarrow$ | $ cE @95 \downarrow$ | samp. time    |
|--------------------------------------|------------------|-----------------------|-----------------------|----------------------|----------------------|----------------------|---------------|
| Treeffuser-published                 | 0.670            | 1.242                 | 0.40                  | 0.117                | 0.044                | 0.023                | 221.35 s      |
| Treeffuser-score <sup>+</sup> (ours) | <b>0.681</b>     | 1.131                 | <b>0.62</b>           | 0.069                | <b>0.020</b>         | <b>0.015</b>         | 34.12 s       |
| Treeffuser-FM (ours)                 | 0.680            | <b>1.117</b>          | 0.46                  | <b>0.064</b>         | 0.028                | 0.024                | 14.41 s       |
| QReg-LightGBM                        | 0.668            | 1.459                 | 0.34                  | 0.066                | 0.041                | 0.044                | 0.30 s        |
| CatBoost-unc.                        | 0.638            | 1.409                 | 0.28                  | 0.093                | 0.078                | 0.066                | <b>0.01 s</b> |
| NGBoost                              | 0.601            | 1.550                 | 0.16                  | 0.119                | 0.114                | 0.101                | 0.04 s        |
| Deep ensemble                        | 0.654            | 1.519                 | 0.26                  | 0.090                | 0.032                | 0.018                | 0.03 s        |
| iBUG                                 | 0.649            | 1.471                 | 0.36                  | 0.111                | 0.051                | 0.065                | 2.89 s        |
| CARD-style diffusion                 | 0.631            | 1.632                 | 0.24                  | 0.086                | 0.057                | 0.043                | 42.84 s       |

Table 2: Per-dataset CRPSS (higher is better) and post-split training rows.

| Dataset       | Train rows | Published    | Score <sup>+</sup> | FM           |
|---------------|------------|--------------|--------------------|--------------|
| yacht         | 256        | 0.941        | <b>0.962</b>       | <b>0.962</b> |
| diabetes      | 368        | 0.201        | <b>0.241</b>       | 0.226        |
| energy        | 640        | 0.956        | 0.962              | <b>0.964</b> |
| concrete      | 858        | 0.740        | <b>0.774</b>       | 0.771        |
| wine          | 5414       | <b>0.346</b> | 0.305              | 0.307        |
| kin8nm        | 6826       | 0.555        | 0.617              | <b>0.631</b> |
| power         | 7973       | 0.829        | 0.839              | <b>0.843</b> |
| naval         | 9945       | 0.950        | 0.949              | <b>0.951</b> |
| cali. housing | 17200      | <b>0.687</b> | 0.676              | 0.674        |
| protein       | 38108      | <b>0.494</b> | 0.480              | 0.472        |

across deep ensembles, CARD-style diffusion, and quantile-regression LightGBM. The Treeffuser aggregate advantage is directional rather than conventionally significant: paired Wilcoxon tests give one-sided  $p = 0.080$  for score<sup>+</sup> over published and  $p = 0.065$  for FM over published; FM versus score<sup>+</sup> remains indistinguishable ( $p = 0.922$ , two-sided).

### 3.2 Per-Dataset CRPSS

Table 2 shows the Treeffuser per-dataset CRPSS. Score<sup>+</sup> is best on concrete and diabetes, FM on energy, kin8nm, naval, power, and yacht, and the published baseline on wine and the two largest datasets. Thus the tuned comparison exposes a Pareto surface: score<sup>+</sup> is usually the calibrated choice, FM the fast normalized-CRPS choice, and the published SDE remains competitive when full SDE sampling is affordable.

**Design diagnostics.** Fixed diagnostics explain the bundles but are not separately tuned factorial comparisons. For FM, the path sweep favours VP under matched residualizer-C and a 5-step Heun ODE; stochastic samplers do not Pareto-dominate  $\varepsilon = 0$  on aggregate CRPS/coverage/latency, though they can improve high-IQR tail coverage (Appendix J); and transferred score-side tricks are neutral or worse. This is why the recommendation is deterministic ODE for aggregate CRPS/latency, with stochasticity left as a conditional-calibration knob. Appendix C maps tuned headline claims versus fixed diagnostics.

## 4 Discussion

Two findings deserve emphasis. First, the headline rows compare tuned bundled recipes, while Appendix C scopes the fixed diagnostics. Because score<sup>+</sup> and FM share the same LightGBM tuning surface, the FM result is not implicitly absorbing tuning-budget improvements that belong to the score track. Second, per-dataset tuning turns the headline into a tradeoff: score<sup>+</sup> is the strongest calibrated Treeffuser row, whereas FM is strongest on normalized CRPS and latency. The common mechanism is regression conditioning for tree ensembles: EDM preconditioning stabilises the VE score path’s inputs and denoising residual, while VP-FM avoids much of the same conditioning problem through

path geometry. This is not a single target-scaling story. Log- $\sigma$  sampling controls where tree capacity is spent, the log- $\sigma$  feature exposes the noise regime to histogram splits, residualization removes predictable conditional location, and Heun ODE sampling is an inference-side efficiency gain. The large-dataset inversion argues against a blanket “SDE is worse” interpretation: per-IQR-bin diagnostics show that stochasticity can improve high-uncertainty tail coverage, while aggregate metrics penalise broader intervals, worse CRPS, and higher latency.

**Limitations.** All displayed families are tuned per dataset, but this is still finite-budget model selection with unequal search spaces and per-trial costs: the comparison is equal finite-trial budget rather than wall-clock-normalized model selection. We include six probabilistic baselines but not distributional random forests [Cevic et al., 2022]; per-dataset gaps, especially on large datasets, should be read as descriptive rather than high-powered significance estimates. The residualizer is also selected from fixed diagnostics rather than automated jointly with the diffusion objective. Choosing between score<sup>+</sup>, FM, and stochastic samplers should ultimately depend on the user’s calibration and latency target, not aggregate CRPS alone.

**Future work.** Useful extensions include wall-clock-normalized tuning, selection criteria for stochastic samplers when conditional coverage matters, multi-output regression, and automatic residualizer selection.

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## 204 A Theoretical Derivations and Design Consequences

205 **Lemma 1 (Wronskian velocity-to-score identity)** *For a differentiable Gaussian path  $y_t =$   
206  $\alpha(t)y_0 + \beta(t)z$  with conditional velocity  $u_t = \alpha'(t)y_0 + \beta'(t)z$ , nonzero Wronskian  $W(t) =$   
207  $\alpha(t)\beta'(t) - \alpha'(t)\beta(t)$ , and  $\beta(t) \neq 0$  (away from the data endpoint), the population velocity  
208  $v(y_t, x, t) = \mathbb{E}[u_t \mid y_t, x, t]$  implies the conditional score*

$$\nabla \log p_t(y_t \mid x) = \frac{\alpha'(t)y_t - \alpha(t)v(y_t, x, t)}{W(t)\beta(t)}.$$

209 *Proof.* Solving the two linear equations for  $z$  gives

$$z = \frac{\alpha(t)u_t - \alpha'(t)y_t}{\alpha(t)\beta'(t) - \alpha'(t)\beta(t)} = \frac{\alpha(t)u_t - \alpha'(t)y_t}{W(t)}.$$

210 Because  $p(y_t \mid y_0, x) = \mathcal{N}(\alpha y_0, \beta^2 I)$ , the conditional score is  $-z/\beta(t)$ . Fisher’s identity gives  
211  $\nabla \log p_t(y_t \mid x) = \mathbb{E}[-z/\beta(t) \mid y_t, x, t]$ . Substituting the expression for  $z$  and using that  $y_t$  is fixed  
212 under the conditional expectation yields the claim. Replacing  $v$  by  $v_\theta$  gives Eq. (2).  $\square$

213 At a fixed  $t$  with  $\alpha(t) \neq 0$ , the same identity inverts to  $v(y_t, x, t) = \alpha'(t)y_t/\alpha(t) -$   
214  $W(t)\beta(t)\nabla \log p_t(y_t \mid x)/\alpha(t)$ . Thus score and velocity have the same conditional structure as  
215 functions of  $y_t$  at that time; the practical differences studied here come from path-dependent scale  
216 variation across  $t$ , feature geometry, and sampler dynamics rather than an intrinsic fixed- $t$  smoothness  
217 advantage.

218 **Proposition 1 (Path-induced feature variance)** *Let  $y_0, z$  be independent with  $\text{Var}(y_0) =$   
219  $\text{Var}(z) = I$  after standardization, and let  $y_t = \alpha(t)y_0 + \beta(t)z$ . Then  $\text{Var}(y_t) = \alpha(t)^2 + \beta(t)^2$ . For  
220 the three paths used in this paper:*

- 221 • Linear ( $\alpha = 1 - t, \beta = t$ ):  $\text{Var}(y_t) = (1 - t)^2 + t^2$ , with minimum  $1/2$  at  $t = 1/2$ .
- 222 • Trigonometric ( $\alpha = \cos(\pi t/2), \beta = \sin(\pi t/2)$ ):  $\text{Var}(y_t) = 1$  for all  $t$ .
- 223 • VP ( $\alpha^2(t) = e^{-T(t)}, \beta^2(t) = 1 - e^{-T(t)}$ ):  $\text{Var}(y_t) = 1$  for all  $t$ .

224 *Proof.* Independence and unit variance give  $\text{Var}(y_t) = \alpha^2 \text{Var}(y_0) + \beta^2 \text{Var}(z) = \alpha^2 + \beta^2$ . Substi-  
225 tuting the three path definitions yields the listed expressions; the trig and VP identities follow from  
226  $\cos^2 + \sin^2 = 1$  and from  $\alpha^2 + \beta^2 = e^{-T} + (1 - e^{-T}) = 1$ .  $\square$

227 **Consequence for tree features.** Proposition 1 formalises why trig and VP paths keep the tree  
228 input near unit scale across the entire path, whereas the linear path compresses to half-variance at  
229  $t = 1/2$ . For histogram-split trees this matters because the empirical distribution of  $y_t$  determines  
230 both candidate split locations and where leaf resolution is spent.

231 **Proposition 2 (Shared histogram-bin approximation)** *Consider one scalar noised-response fea-  
232 ture and a finite histogram partition  $\mathcal{P}_{\bar{p}} = \{I_1, \dots, I_B\}$  whose bins have equal mass under the  
233 pooled path marginal*

$$\bar{p}(y) = \int p(t) p_t(y) dt.$$

234 Let  $\mathcal{H}(\mathcal{P}_{\bar{p}})$  be the functions that are constant on each  $I_b$ , and let  $f_t(y) = \mathbb{E}[T_t \mid y_t = y, x, t]$  denote  
 235 the scalar regression function for a score, denoising, noise, or velocity target at a selected time  $t$  with  
 236 covariate value  $x$  fixed. Then

$$\inf_{g \in \mathcal{H}(\mathcal{P}_{\bar{p}})} \mathbb{E}_{p_t} [(f_t(y_t) - g(y_t))^2] \leq \sum_{b=1}^B p_t(I_b) \text{osc}_{I_b}(f_t)^2,$$

237 where  $\text{osc}_{I_b}(f_t) = \sup_{y, y' \in I_b} |f_t(y) - f_t(y')|$ . If  $f_t$  is  $L_t$ -Lipschitz on the relevant bins, the right-  
 238 hand side is at most  $L_t^2 \sum_b p_t(I_b) |I_b|^2$ .

239 *Proof.* The best piecewise-constant approximation on a fixed bin partition cannot be worse than any  
 240 particular constant choice on each bin. Choosing any value between the infimum and supremum of  
 241  $f_t$  on  $I_b$  gives pointwise squared error at most  $\text{osc}_{I_b}(f_t)^2$  on that bin, and integration under  $p_t$  gives  
 242 the first bound. The Lipschitz bound follows from  $\text{osc}_{I_b}(f_t) \leq L_t |I_b|$ .  $\square$

243 **Shared-bin mismatch.** Proposition 2 is a single-feature proxy for the split candidates available to a  
 244 histogram tree, not a theory of the full boosted LightGBM ensemble. Its useful comparison is against  
 245 an oracle partition that can choose  $B$  bins separately for each  $p_t$ . The excess cost of using one pooled  
 246 partition,

$$\Delta_t(B) = \inf_{g \in \mathcal{H}(\mathcal{P}_{\bar{p}})} \mathbb{E}_{p_t} [(f_t(y_t) - g(y_t))^2] - \inf_{g \in \mathcal{H}(\mathcal{P}_t^*)} \mathbb{E}_{p_t} [(f_t(y_t) - g(y_t))^2],$$

247 where  $\mathcal{P}_t^*$  is the best  $B$ -bin partition for time  $t$ , captures the finite shared-bin cost. VP and trigonomet-  
 248 ric paths keep the  $p_t$  family near unit scale, so pooled bins are close to useful per-time bins. VE paths  
 249 mix data-scale low-noise features with expanded high-noise features; a single pooled histogram must  
 250 compromise across regimes. EDM input preconditioning and VP-FM both reduce this mismatch.  
 251 Figure 1(b) illustrates the same effect empirically: the published-style VE target is stable, but the  
 252 unpreconditioned tree input expands across noise levels, whereas EDM and VP-FM keep the feature  
 253 scale closer to a shared-bin regime. This is a comparative statement about shared finite bins, not a  
 254 claim that the fixed-time VE regression function must become rough: for example,  $\mathbb{E}[z \mid y_t, x, t]$   
 255 can become smoother as  $\sigma$  grows even while the pooled histogram becomes a poorer substitute for  
 256 per-time resolution.

257 **Residualization removes predictable location.** Residualization is an exact reparameterization in  
 258 the infinite-capacity limit: for any fixed  $\hat{\mu}(x)$ , modelling  $r = y - \hat{\mu}(x)$  exactly and returning  $y =$   
 259  $r + \hat{\mu}(x)$  recovers the same conditional distribution. Its empirical value is finite-model conditioning.  
 260 If  $\mu(x) = \mathbb{E}[y \mid x]$  and  $r = y - \mu(x)$ , then  $\mathbb{E}[\|r\|^2 \mid x] = \text{tr Var}(y \mid x) \leq \mathbb{E}[\|y\|^2 \mid x]$ , so diffusion  
 261 or FM on residuals removes a deterministic location component that would otherwise be present  
 262 at every noise level. The inequality also explains the large-dataset caveat: a fixed residualizer can  
 263 become a bottleneck once the inner score/velocity model has enough data to represent the mean  
 264 directly.

265 **Proposition 3 (Endpoint score-error amplification)** Let  $v_\theta(y_t, x, t) = v(y_t, x, t) + \delta(y_t, x, t)$  be  
 266 an approximate velocity with pointwise error  $\delta$ , and let  $\hat{s}_\theta, s$  denote the implied and true scores from  
 267 Lemma 1. Then

$$\hat{s}_\theta(y_t, x, t) - s(y_t, x, t) = -\frac{\alpha(t)}{W(t)\beta(t)} \delta(y_t, x, t),$$

268 so the pointwise score-error gain is  $|\alpha(t)/(W(t)\beta(t))|$ . For all three paths considered here  $W(t) \rightarrow$   
 269  $\alpha(0)\beta'(0) > 0$  is bounded near  $t = 0$  while  $\beta(t) \rightarrow 0$ , so the gain diverges at the data endpoint.

270 *Proof.* Eq. (2) is linear in  $v$ , so

$$\hat{s}_\theta - s = \frac{\alpha'(t)y_t - \alpha(t)(v + \delta) - (\alpha'(t)y_t - \alpha(t)v)}{W(t)\beta(t)} = -\frac{\alpha(t)\delta}{W(t)\beta(t)}. \quad \square$$

271 **Endpoint amplification.** Proposition 3 therefore predicts that any stochasticity schedule with  
 272  $\varepsilon(t) \not\rightarrow 0$  as  $t \rightarrow 0$  injects high-gain approximate score into the SDE drift. In Eq. (3), the recovered-  
 273 score contribution to the drift error is

$$\frac{\varepsilon(t)^2}{2}(\hat{s}_\theta - s) = -\frac{\varepsilon(t)^2\alpha(t)}{2W(t)\beta(t)}\delta.$$

274 Near the data endpoint for the paths used here,  $\beta(t) \asymp t$ ,  $\alpha(t) \asymp 1$ , and  $W(t)$  stays bounded away  
 275 from zero, so this term scales like  $\varepsilon(t)^2 \delta/t$ . Constant stochasticity amplifies endpoint velocity error,  
 276  $\varepsilon(t) \asymp \sqrt{t}$  leaves a non-vanishing endpoint drift-error scale,  $\varepsilon(t) = ct$  makes this recovered-score  
 277 contribution vanish, and  $\varepsilon \equiv 0$  removes it entirely. This is why  $\varepsilon(t) = ct$  and the deterministic corner  
 278 are the two natural schedules in §2.2.

279 **Proposition 4 (Population equivalence on the VE path)** *For a VE path  $y_t = y_0 + \beta(t)z$  with*  
 280  *$\beta'(t) > 0$ , training a normalized velocity model  $\nu_\theta$  against  $u_t/\beta'(t) = z$  implies the same population*  
 281 *score as VE noise prediction and VE denoising/EDM score reconstruction:*

$$s_t(y_t, x) = -\frac{\mathbb{E}[z \mid y_t, x, t]}{\beta(t)} = \frac{\mathbb{E}[y_0 \mid y_t, x, t] - y_t}{\beta(t)^2}.$$

282 *Proof.* For VE,  $\alpha = 1$ ,  $\alpha' = 0$ ,  $u_t = \beta'(t)z$ , and  $W = \beta'(t)$ . Lemma 1 gives  $s_t = -\mathbb{E}[u_t \mid$   
 283  $y_t, x, t]/(\beta(t)\beta'(t)) = -\mathbb{E}[z \mid y_t, x, t]/\beta(t)$ . Since  $y_t = y_0 + \beta(t)z$ ,  $\mathbb{E}[y_0 \mid y_t, x, t] = y_t - \beta(t)\mathbb{E}[z \mid$   
 284  $y_t, x, t]$ , which gives the denoising form. Noise prediction estimates  $-\mathbb{E}[z \mid y_t, x, t]$  and EDM  
 285 reconstructs the same population denoiser after its skip/output/input preconditioning.  $\square$

286 Proposition 4 is a population identity, not a claim that the finite LightGBM problems are identical.  
 287 EDM fits the preconditioned denoising residual  $F = (y_0 - c_{\text{skip}}y_t)/c_{\text{out}}$  and reconstructs  $D_\theta =$   
 288  $c_{\text{skip}}y_t + c_{\text{out}}F_\theta$ , whereas normalized VE-FM would fit  $z$  directly. They imply the same score  
 289 at the optimum but present different targets and feature scalings to LightGBM. This is the useful  
 290 interpretation of the score<sup>+</sup>-FM tie: both improve the conditioning of the regression problem seen  
 291 by the tree ensemble, EDM by preconditioning the VE score objective and VP-FM by choosing a  
 292 better-conditioned probability path.

Table 3: Regression-conditioning view of the main design choices. Here  $y_0$  and  $z$  are standardized.  
 “Conditioning issue” refers to the supervised problem fit by LightGBM, not to the population score  
 identity.

| Variant                   | Regressed object                            | Feature scale                         | Conditioning issue / fix                                                   |
|---------------------------|---------------------------------------------|---------------------------------------|----------------------------------------------------------------------------|
| VE noise prediction       | $-z$                                        | $y_0 + \sigma z$ grows with $\sigma$  | Unit target, but heteroscedastic input and $1/\sigma$ score amplification. |
| VE EDM score <sup>+</sup> | $(y_0 - c_{\text{skip}}y_t)/c_{\text{out}}$ | $c_{\text{in}}y_t$ is near unit scale | Input and denoising residual are explicitly preconditioned.                |
| Raw VE-FM                 | $\beta'(t)z$                                | $y_0 + \beta z$ grows with $\beta$    | Velocity target inherits the schedule derivative.                          |
| Normalized VE-FM          | $z$                                         | $y_0 + \beta z$ unless also scaled    | Population-equivalent to noise prediction after known rescaling.           |
| VP-FM                     | $\alpha'y_0 + \beta'z$                      | $\alpha^2 + \beta^2 \approx 1$        | Path keeps tree features near the standardized data scale.                 |

## 293 B Datasets and Protocol Details

294 **Sources.** The ten UCI benchmarks follow the canonical Treeffuser protocol [Beltrán-Velez  
 295 et al., 2024]: `yacht`, `concrete`, `energy`, `wine`, `kin8nm`, `naval`, `power_plant`, `protein` from  
 296 the UCI repository; `california_housing` from scikit-learn’s `fetch_california_housing`;  
 297 and `diabetes` from scikit-learn’s `load_diabetes`. The headline results use the six-fold tun-  
 298 ing/evaluation protocol from §3; Table 2 reports the post-split training rows for each evaluation  
 299 fold. Fixed-configuration appendix diagnostics instead use the train/test sizes declared in their  
 300 YAML configs, sometimes with large datasets subsampled for sweep cost. The only preprocessing  
 301 is StandardScaler standardisation of features and target, fit on the training split and applied to the  
 302 corresponding test split. `wine` is the union of `wine-quality-red` and `wine-quality-white` with  
 303 a binary `color` indicator.

304 **Seed policy.** The fold-tuned headline uses the deterministic six-fold manifest with master seed 0.  
 305 Fixed diagnostic sweeps use the seed lists declared in their configs (usually 0–2 or 0–9), with three  
 306 independent streams: `data_seed` (offset 0) for the train/test split; `model_seed` (offset 10,000) for  
 307 the LightGBM regressors; `sampler_seed` (offset 20,000) for stochastic samplers. Variance reduction  
 308 across seeds therefore reflects only sampling-and-split variability, not boosting initialisation.



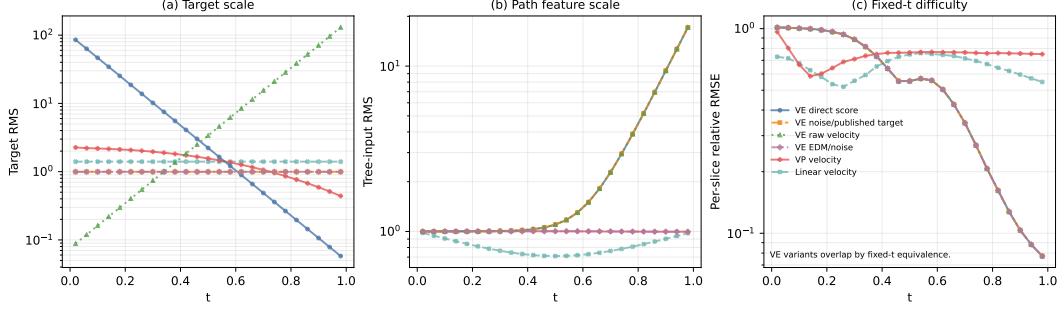


Figure 1: Toy preconditioning diagnostic on a one-dimensional heteroscedastic mixture, generated by `pixi run toy-geometry`. Panel (a) shows that direct VE score prediction and raw VE-FM have target scales that vary by orders of magnitude across  $t$ , whereas the published-style VE noise target, EDM/noise prediction, and linear velocity have unit target scale. Panel (b) separates target scaling from input geometry: the published-style VE target is stable but the tree input  $y_t = y_0 + \sigma z$  still expands strongly, while EDM rescales the input and VP-FM keeps the path feature near standardized scale. Panel (c) fits a fresh LightGBM at each fixed  $t$ ; the VE curves overlap up to numerical noise, confirming that the score/velocity distinction is not intrinsic fixed- $t$  smoothness but cross- $t$  conditioning.

**Common training settings.** In fixed diagnostic sweeps, unless noted, Treeffuser variants use  $n_{\text{estimators}} = 3000$ , early stopping after 50 rounds, learning rate 0.1,  $n_{\text{repeats}} = 30$  (each training point is reused with 30 fresh  $t$  samples), and the per-output-dimension LightGBM fitter. In the headline protocol, these and other LightGBM hyperparameters are selected per dataset from the tuning spaces and stored under `benchmarks/results/tuning/best_params/`. Inference draws 200 samples per test point.

**Tuning budget.** For each dataset/model-family pair, Optuna targets 25 finite trials on the fold-0 train/validation split, with at most twice as many total attempts to allow failed or non-finite trials. The tuning objective is validation CRPS computed from 100 predictive Monte Carlo samples per validation point; this is the number of generated response samples used to estimate CRPS, not a subsample of validation observations. Final evaluation refits the selected configuration on each held-out evaluation fold and draws 200 samples per test point. The three Treeffuser rows share the same six-dimensional LightGBM search surface: `n_estimators`, `learning_rate`, `num_leaves`, `max_depth`, `min_child_samples`, and `subsample`; method-defining choices such as objective, residualization, probability path, and sampler are fixed by row. External baselines are also tuned for 25 finite trials, but their search spaces and per-trial costs differ, so the comparison should be read as equal finite-trial budget rather than wall-clock-normalized model selection. The persisted YAML files record `n_trials_target`, `n_trials_finite`, and `n_trials_failed`. Table 4 summarises the per-family tunable surfaces.

**Appendix calibration metrics.** Appendix tables use  $q$ -MACE for sample-rank calibration: for each held-out response, we compute the fraction of predictive samples above the observed value, sort these ranks, and average their absolute deviation from an evenly spaced uniform grid. Lower  $q$ -MACE therefore means the sample predictive distributions have ranks closer to uniform, analogous to the PIT-KS metric in the main table but reported as an error magnitude.

**Sampler details.** TREEFFUSER-PUBLISHED uses the VESDE reverse-time SDE with the Euler-Maruyama sampler at  $n_{\text{steps}} = 50$ , matching the published code path. TREEFFUSER-SCORE<sup>+</sup> uses the Heun probability-flow ODE at  $n_{\text{steps}} = 25$  steps under the EDM-style parametrization with  $\sigma_{\min} = 0.01$ ,  $\sigma_{\max} = 20$ . TREEFFUSER-FM uses the Heun ODE at  $n_{\text{steps}} = 5$  on the VP path with  $\beta_{\min} = 0.1$ ,  $\beta_{\max} = 20$ .

Table 4: Tunable hyperparameter ranges per model family (25 Optuna trials each). Log = log-uniform sampling. Fixed method-defining parameters are not listed.

| Family                 | Tunable parameter | Range               | Scale       |
|------------------------|-------------------|---------------------|-------------|
| Treefuser (all 3 rows) | n_estimators      | 200–3000            | log         |
|                        | learning_rate     | 0.001–0.3           | log         |
|                        | num_leaves        | 15–255              | linear      |
|                        | max_depth         | {−1, 4, 6, 8, 10}   | categorical |
|                        | min_child_samples | 5–100               | linear      |
|                        | subsample         | 0.5–1.0             | linear      |
| NGBoost                | n_estimators      | 200–3000            | log         |
|                        | learning_rate     | 0.001–0.3           | log         |
| iBUG                   | k                 | 10–200              | log         |
|                        | n_estimators      | 200–2000            | log         |
|                        | learning_rate     | 0.001–0.3           | log         |
|                        | max_depth         | 3–10                | linear      |
| QReg-LightGBM          | quantile_count    | 24–49               | linear      |
|                        | n_estimators      | 50–500              | log         |
|                        | learning_rate     | 0.02–0.2            | log         |
|                        | num_leaves        | 15–127              | linear      |
| CatBoost               | iterations        | 200–3000            | log         |
|                        | learning_rate     | 0.001–0.3           | log         |
|                        | depth             | 4–10                | linear      |
| Deep ensemble          | hidden_size       | {64, 128, 256, 512} | categorical |
|                        | n_layers          | 2–5                 | linear      |
|                        | learning_rate     | 0.0001–0.01         | log         |
|                        | max_epochs        | {100, 200, 400}     | categorical |
| CARD-style             | hidden_size       | {64, 128, 256}      | categorical |
|                        | n_layers          | 2–5                 | linear      |
|                        | learning_rate     | 0.0001–0.01         | log         |
|                        | max_epochs        | {200, 400}          | categorical |
|                        | diffusion_epochs  | {200, 400}          | categorical |
|                        | n_steps           | {50, 100, 200}      | categorical |

## C Claim Scope and Diagnostic Cross-References

Unless otherwise stated, appendix sweeps are fixed-configuration diagnostics used to isolate modelling choices; they are not per-dataset tuned headline comparisons. The experimental pipeline is: fixed diagnostics identify plausible design choices; those choices define the frozen PUBLISHED, SCORE<sup>+</sup>, and FM model families; the frozen families are then tuned per dataset under the fold-0 protocol; and only that tuned protocol supports the headline aggregate performance claims.

Table 5: Scope of the empirical claims. “Tuned” means per-dataset Optuna selection on fold 0 before evaluation on folds 1–5; “fixed” means shared diagnostic settings declared in the corresponding YAML configs.

| Claim                                                                                                                 | Evidence                                                 | Tuning status     | Role in the paper                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|
| SCORE <sup>+</sup> and FM improve the Treefuser tradeoff surface over the published row                               | Table 1 and Table 2                                      | Tuned per dataset | Confirmatory performance comparison among frozen model families                                                    |
| SCORE <sup>+</sup> is the most calibrated Treefuser row under this benchmark protocol                                 | Table 1                                                  | Tuned per dataset | Headline aggregate calibration claim                                                                               |
| VP-FM is the preferred FM path among the tested paths                                                                 | Table 12                                                 | Fixed diagnostic  | Mechanism and model-selection evidence; not a claim that VP would win every independently tuned path comparison    |
| Deterministic FM is the recommended aggregate CRPS/latency sampler                                                    | Table 11 with Table 15                                   | Fixed diagnostic  | Sampler diagnostic; stochasticity remains a conditional-calibration knob                                           |
| Residualizer-C is a reasonable headline residualizer choice                                                           | Tables 6 and 7                                           | Fixed diagnostic  | Design choice for the frozen model families; automatic residualizer selection is left as future work               |
| Log- $\sigma$ sampling, explicit noise-level features, and EDM preconditioning motivate the SCORE <sup>+</sup> bundle | Toy diagnostic and development sweeps listed in Table 16 | Fixed diagnostic  | Mechanism evidence for the bundle; the quantitative performance claim is the tuned combined SCORE <sup>+</sup> row |

The four score-side modifications of §2.1 were validated by smoke and synthetic-core sweeps before being combined in the SCORE<sup>+</sup> variant. The residualizer-C configuration that the headline uses was selected by the residualizer sweep summarised in Appendix D. Log-sigma time sampling, the log-sigma noise feature, and EDM preconditioning were locked in by the `log_sigma_sweep`, `synthetic_core`, and `pf_ode_sweep` development runs. We treat those runs as model-selection evidence rather than as separately reported tuned comparisons; the quantitative claim in the paper is the combined SCORE<sup>+</sup> row in Table 1, where all four choices are evaluated together on the full ten-dataset protocol.

## D Residualizer Configuration

### D.1 Off vs. Mean Residualization (FM)

We isolate the contribution of mean residualization on the FM side at matched configuration (VP path, uniform  $t$ , uniform loss weighting, residualizer-C extra parameters) using a ten-dataset mixed synthetic/real diagnostic suite and ten seeds.

Table 6: FM with mean residualization vs. no residualization (off). Paired differences (mean – off, 100 paired observations); negative is better for all metrics shown.

| Sampler        | $\Delta$ CRPS | $\Delta q$ -MACE | $\Delta$ KS stat | $\Delta  cE @90$ | paired $t$ on $\Delta$ CRPS |
|----------------|---------------|------------------|------------------|------------------|-----------------------------|
| ODE @ 5 steps  | −0.23         | −0.022           | −0.034           | −0.032           | −4.30                       |
| SDE @ 25 steps | −0.05         | +0.001           | +0.003           | −0.001           | −2.91                       |

The ODE-5 advantage is large and broadly significant across calibration metrics (81/100 paired wins on  $q$ -MACE and 81/100 on  $|cE|@90$ ). At SDE-25 the gap on calibration metrics is null (paired  $t < 1$ ); the aggregate CRPS gain is driven by two real-data outliers (yacht +54% relative, concrete +15%) while six of ten datasets actually favour the unresidualized baseline. The mechanism is consistent with the few-step ODE being unable to absorb the conditional mean itself when the velocity net has  $\leq 5$  Heun updates; with 25 SDE updates the velocity model recovers calibration on its own. Cost: mean residualization adds  $\approx 12$  s of fit time on these sizes ( $\sim 8\times$  vs. no residualization), so the trade is most attractive in low-step regimes.

### D.2 Residualizer Variants A–E (FM side)

Five residualizer configurations were swept on a four-dataset, three-seed sub-suite (california\_housing, diabetes, energy, protein-5k subsample) under VP-FM-ODE at 5 steps; see Table 7.

Table 7: FM residualizer diagnostic. “Conf. leaves” is the LightGBM `num_leaves`; “ $n_{\text{est}}^{(r)}$ ” is the residualizer’s `n_estimators`; “ES” is early-stopping rounds in the residualizer (off means no ES).

| Variant | Description        | Leaves | $n_{\text{est}}^{(r)}$ | ES  | CRPSS $\uparrow$ | $q$ -MACE $\downarrow$ | $ cE @95 \downarrow$ | fit s |
|---------|--------------------|--------|------------------------|-----|------------------|------------------------|----------------------|-------|
| A       | current-baseline   | 31     | 100                    | off | 0.521            | 0.028                  | 0.017                | 1.82  |
| B       | regularised        | 31     | 100                    | 30  | 0.509            | 0.042                  | 0.018                | 2.78  |
| C       | high capacity      | 63     | 300                    | off | <b>0.523</b>     | 0.039                  | 0.011                | 8.03  |
| D       | ES moderate        | 63     | 500                    | 30  | 0.516            | 0.033                  | 0.021                | 5.91  |
| E       | ES + high capacity | 63     | 2000                   | 30  | 0.522            | 0.034                  | <b>0.010</b>         | 9.81  |

Within this fixed diagnostic, configurations C and E sit on the calibration–latency Pareto front. C is the headline choice (best CRPSS,  $1.2\times$  faster than E); E edges C on tail coverage ( $|cE|@95$  0.010 vs. 0.011). A separate full-protein diagnostic shows E reclaiming the lead on protein (CRPSS 0.476 vs. C’s 0.449), reinforcing the case that the optimal residualizer is dataset-dependent.

### D.3 Scale-Aware Residualization

The `mean_scale` residualizer additionally divides residuals by a cross-validated conditional-std estimate  $\hat{\sigma}(x)$ . On the ten-dataset mixed synthetic/real diagnostic suite with ten seeds:

Table 8: Mean vs. mean-scale residualization on FM at matched configuration. Higher CRPSS is better; lower CRPS, DSS,  $q$ -MACE are better.

| Variant               | CRPSS $\uparrow$ | CRPS $\downarrow$ | DSS $\downarrow$ | $q$ -MACE $\downarrow$ |
|-----------------------|------------------|-------------------|------------------|------------------------|
| VP-FM-ODE, mean       | <b>0.519</b>     | <b>0.652</b>      | +0.314           | <b>0.031</b>           |
| VP-FM-ODE, mean-scale | 0.489            | 0.724             | +0.665           | 0.038                  |
| VP-FM-SDE, mean       | <b>0.517</b>     | <b>0.657</b>      | +0.380           | <b>0.040</b>           |
| VP-FM-SDE, mean-scale | 0.475            | 0.750             | +0.676           | 0.049                  |

Scale-aware residualization is uniformly worse: +11% CRPS on the ODE path, +14% on the SDE path, with DSS more than doubling. The per- $x$  scale estimator is itself noisy in these finite-sample regimes; dividing by it injects more variance into the velocity target than it removes. `mean` is therefore the only residualizer carried into the headline.

## E Time Sampling and Loss Weighting

### E.1 Time Sampling

Table 9: FM time-sampling sweep on fixed UCI train/test splits, 10 datasets  $\times$  3 seeds, matched residualizer-C, VP path. “uniform” is the headline choice with an endpoint anchor at  $t = 1$  at probability 0.05.

| Variant                            | CRPSS $\uparrow$ | DSS $\downarrow$ | $q$ -MACE $\downarrow$ | KS $p > .05 \uparrow$ | samp s |
|------------------------------------|------------------|------------------|------------------------|-----------------------|--------|
| VP-FM-ODE uniform                  | <b>0.6564</b>    | −0.154           | 0.040                  | 0.47                  | 3.63   |
| VP-FM-ODE uniform anchor16         | 0.6560           | −0.171           | 0.040                  | 0.53                  | 3.84   |
| VP-FM-ODE logbeta                  | 0.6556           | −0.175           | <b>0.035</b>           | <b>0.77</b>           | 3.74   |
| VP-FM-ODE logSNR                   | 0.6538           | −0.135           | 0.042                  | 0.40                  | 3.71   |
| score <sup>+</sup> uniform- $t$    | 0.6556           | −0.210           | 0.038                  | 0.50                  | 6.95   |
| score <sup>+</sup> log- $\sigma t$ | <b>0.6565</b>    | −0.169           | 0.038                  | <b>0.70</b>           | 6.58   |

For FM, log-beta and log-SNR sampling do not move CRPS by more than 0.4% relative to uniform with endpoint anchor; the smoother density shape gains +5% on KS pass rate and −13% on  $q$ -MACE but the effect is within seed variance for CRPSS. The headline keeps uniform- $t$  for parsimony; log-beta is a viable substitute when KS uniformity is the priority. For the score model, log-sigma sampling clearly helps PIT uniformity (KS pass 0.50  $\rightarrow$  0.70) at the cost of slightly worse DSS; we keep log-sigma in the SCORE<sup>+</sup> headline because the KS gain is the more interpretable calibration win.

### E.2 Loss Weighting (min-SNR- $\gamma$ )

Table 10: Min-SNR- $\gamma$  loss weighting on FM, fixed UCI train/test splits, 10 datasets  $\times$  3 seeds.  $\gamma = 5$  collapses to uniform on most of the support;  $\gamma = 1$  amplifies the late-time loss most aggressively.

| Variant                | CRPSS $\uparrow$ | CRPS $\downarrow$ | DSS $\downarrow$ | $q$ -MACE $\downarrow$ | KS $p > .05 \uparrow$ |
|------------------------|------------------|-------------------|------------------|------------------------|-----------------------|
| VP-FM-ODE uniform- $w$ | <b>0.6564</b>    | <b>4.097</b>      | −0.159           | 0.040                  | 0.47                  |
| VP-FM-ODE min-SNR-5    | 0.6564           | 4.119             | −0.167           | <b>0.037</b>           | <b>0.60</b>           |
| VP-FM-ODE min-SNR-1    | 0.6516           | 4.223             | −0.104           | 0.044                  | 0.37                  |
| VP-FM-SDE uniform- $w$ | <b>0.6540</b>    | <b>4.114</b>      | −0.080           | 0.051                  | 0.30                  |
| VP-FM-SDE min-SNR-5    | 0.6538           | 4.138             | −0.085           | 0.050                  | 0.30                  |
| VP-FM-SDE min-SNR-1    | 0.6485           | 4.239             | −0.046           | 0.057                  | 0.23                  |

Min-SNR- $\gamma$  is a sharpness-versus-tail trade-off.  $\gamma = 1$  hurts CRPS by 0.8–1.3% across both samplers (Wilcoxon  $p \leq 0.003$  for both);  $\gamma = 5$  is statistically indistinguishable from uniform weighting on CRPS, but improves PIT KS pass rate on the ODE path. We keep uniform weighting in the FM headline.

## F Stochasticity (Velocity-Noise) Sweeps

### F.1 Stochasticity Strength

We sweep velocity stochasticity on the VP path under linear and  $\sqrt{t}$  schedules with residualizer-C, plus one residualizer-E check; results in Table 11 are averages over fixed UCI train/test splits, 10 datasets  $\times$  3 seeds at 25 SDE steps.

Table 11: Stochasticity sweep. The  $\varepsilon = 0$  corner is the deterministic FM recipe (Heun ODE @ 5 steps); SDE rows use 25 steps. Sample time is omitted because the ODE and SDE rows use different benchmark protocols.

| Variant                            | CRPSS $\uparrow$ | CRPS $\downarrow$ | cE @90 $\downarrow$ | cE @95 $\downarrow$ |
|------------------------------------|------------------|-------------------|---------------------|---------------------|
| $\varepsilon = 0$ headline (ODE)   | 0.669            | –                 | <b>0.024</b>        | <b>0.017</b>        |
| $\varepsilon = 0.25$ linear, res-C | 0.656            | 4.099             | 0.028               | 0.015               |
| $\varepsilon = 0.50$ linear, res-C | 0.655            | 4.104             | 0.036               | 0.017               |
| $\varepsilon = 1.00$ linear, res-C | 0.654            | 4.114             | 0.047               | 0.022               |
| $\varepsilon = 0.50$ sqrt, res-C   | 0.655            | 4.101             | 0.035               | 0.017               |
| $\varepsilon = 1.00$ sqrt, res-C   | 0.655            | 4.107             | 0.043               | 0.020               |
| $\varepsilon = 1.00$ linear, res-E | 0.651            | 4.128             | 0.048               | 0.024               |

In this fixed diagnostic sweep, no tested SDE setting Pareto-dominates  $\varepsilon = 0$ . The closest contender ( $\varepsilon = 0.25$  linear) matches |cE|@95 (0.015 vs. 0.017) but is strictly worse on CRPSS and |cE|@90, and pays 25 SDE steps vs. 5 ODE steps.

### F.2 Schedule Shape

The schedule  $\varepsilon(t) = ct$  (linear) is the headline form. The  $\sqrt{t}$  alternative was tested for parity at  $\varepsilon \in \{0.5, 1.0\}$  and produces statistically indistinguishable CRPSS (Table 11). Both schedules vanish at the data endpoint, which prevents the  $1/\beta(t)$  blow-up in the velocity-to-score conversion (Eq. (2)); neither shape helps once  $\varepsilon$  is small.

## G Path Choice

Table 12: Flow-path diagnostic under the 5-step deterministic ODE sampler: linear vs. trigonometric vs. variance-preserving, 8 datasets  $\times$  3 seeds, matched residualizer-C. Lower is better for all columns; bold is best per column.

| Path                     | CRPS $\downarrow$ | $q$ -MACE $\downarrow$ | cE @95 $\downarrow$ | samp s      |
|--------------------------|-------------------|------------------------|---------------------|-------------|
| Linear                   | 4.324             | <b>0.026</b>           | 0.032               | 0.49        |
| Trigonometric            | 4.306             | 0.028                  | <b>0.028</b>        | <b>0.43</b> |
| Variance-preserving (VP) | <b>4.294</b>      | 0.028                  | <b>0.028</b>        | 0.46        |

VP is best on CRPS under the few-step ODE, while trig and VP both reduce 95% coverage error relative to the linear path. The advantage tracks variance preservation: VP and trig satisfy  $\alpha^2 + \beta^2 \approx 1$ , keeping  $y_t$  at roughly unit scale across  $t$  and stabilising the tree regressor’s feature space. The headline uses VP.

## H Probabilistic-Family Baselines

We compare the headline Treeffuser variants against six external probabilistic regressors (Table 13). Hyperparameters for every displayed family are selected per dataset on the same fold-0 tuning split, then evaluated on folds 1–5. The CARD row uses a compact benchmark adapter with the same two-stage conditional-mean plus conditional-diffusion structure as CARD [Han et al., 2022], so conclusions involving CARD should be read as comparisons to this adapter rather than to a fully optimized CARD pipeline.

Table 13: Probabilistic baselines under the tuning/evaluation protocol. rel-CRPS is normalized by the best displayed variant on each dataset. Sample-time excludes fitting. The iBUG DSS is dominated by a single wine fold where the model assigns near-zero predictive variance to a mispredicted test point.

| Model                                | CRPSS $\uparrow$ | rel-CRPS $\downarrow$ | CRPS $\downarrow$ | DSS $\downarrow$   | $q$ -MACE $\downarrow$ | cE @95 $\downarrow$ | samp s      |
|--------------------------------------|------------------|-----------------------|-------------------|--------------------|------------------------|---------------------|-------------|
| Treeffuser-score <sup>+</sup> (ours) | <b>0.681</b>     | 1.131                 | 4.057             | -0.214             | <b>0.028</b>           | <b>0.015</b>        | 34.12       |
| Treeffuser-FM-fast (ours)            | 0.680            | <b>1.117</b>          | 4.121             | <b>-0.218</b>      | 0.034                  | 0.024               | 14.41       |
| Treeffuser-published                 | 0.670            | 1.242                 | 4.279             | 1.358              | 0.044                  | 0.023               | 221.35      |
| Quantile-regression LightGBM         | 0.668            | 1.459                 | 4.026             | 0.016              | 0.033                  | 0.044               | 0.30        |
| Deep ensemble                        | 0.654            | 1.519                 | <b>4.008</b>      | 0.090              | 0.050                  | 0.018               | 0.03        |
| iBUG (XGBoost)                       | 0.649            | 1.471                 | 4.193             | $1.85 \times 10^7$ | 0.069                  | 0.065               | 2.89        |
| CatBoost (uncertainty)               | 0.638            | 1.409                 | 4.045             | 0.272              | 0.045                  | 0.066               | <b>0.01</b> |
| CARD-style diffusion                 | 0.631            | 1.632                 | 4.838             | 0.436              | 0.051                  | 0.043               | 42.84       |
| NGBoost (Gaussian)                   | 0.601            | 1.550                 | 4.299             | 1.957              | 0.052                  | 0.101               | 0.04        |

Under this per-dataset tuning protocol, the Treeffuser variants lead on aggregate CRPSS, rel-CRPS, and coverage metrics. The external baselines are still important: deep ensembles have the best mean raw CRPS, though this scale-sensitive average is less informative about cross-dataset dominance than CRPSS or rel-CRPS, and the per-dataset winners include deep ensembles on diabetes and kin8nm, CARD on naval, and quantile-regression LightGBM on protein. Their aggregate rel-CRPS remains worse, indicating that no single tuned external family transfers as evenly across the benchmark suite as the Treeffuser variants.

## I Extended Per-Dataset Results

Table 14 reports CRPS, |cE|@90, and sample-generation time on each of the ten benchmarks for the three headline Treeffuser variants, averaged over the five evaluation folds. The “best margin” column gives the gap between the winning variant and the second-best Treeffuser variant in raw CRPS. After per-dataset tuning, the margins are tighter: the main result is no longer a few dramatic wins, but a more balanced Pareto surface between published SDE sampling, score-side conditioning, and FM speed.

Table 14: Extended per-dataset Treeffuser results, ordered roughly by evaluation-fold training size. “best margin” is winner-over-second-best in raw CRPS; the bold cell in each block is the per-metric winner on that row. Numbers come from `benchmarks/results/tuning/eval/*.jsonl`.

| Dataset       | best margin            | CRPS $\downarrow$ |                    |                 | cE @90 $\downarrow$ |                    |              | sample s |                    |             |
|---------------|------------------------|-------------------|--------------------|-----------------|---------------------|--------------------|--------------|----------|--------------------|-------------|
|               |                        | Pub.              | Score <sup>+</sup> | FM              | Pub.                | Score <sup>+</sup> | FM           | Pub.     | Score <sup>+</sup> | FM          |
| yacht         | FM +0.4%               | 0.438             | 0.284              | <b>0.283</b>    | 0.092               | <b>0.025</b>       | 0.081        | 0.3      | 0.3                | <b>0.1</b>  |
| diabetes      | Sco <sup>+</sup> +1.9% | 35.682            | <b>34.025</b>      | 34.660          | 0.079               | <b>0.029</b>       | 0.038        | 2.4      | 0.5                | <b>0.2</b>  |
| energy        | FM +5.2%               | 0.252             | 0.214              | <b>0.204</b>    | 0.081               | 0.034              | <b>0.022</b> | 2.9      | 0.8                | <b>0.6</b>  |
| concrete      | Sco <sup>+</sup> +1.5% | 2.457             | <b>2.126</b>       | 2.158           | 0.034               | 0.025              | <b>0.023</b> | 3.0      | 4.8                | <b>0.8</b>  |
| wine          | Pub +5.9%              | <b>0.300</b>      | 0.319              | 0.318           | 0.037               | <b>0.006</b>       | 0.016        | 100.4    | 37.0               | <b>11.9</b> |
| kin8nm        | FM +4.0%               | 0.067             | 0.058              | <b>0.055</b>    | 0.012               | <b>0.004</b>       | 0.023        | 22.0     | 20.2               | <b>9.2</b>  |
| power         | FM +2.4%               | 1.671             | 1.572              | <b>1.535</b>    | 0.013               | <b>0.004</b>       | 0.006        | 14.6     | <b>6.7</b>         | 7.6         |
| naval         | FM +3.2%               | 0.000218          | 0.000220           | <b>0.000211</b> | 0.078               | 0.065              | <b>0.061</b> | 207.4    | 91.2               | <b>29.5</b> |
| cali. housing | Pub +3.6%              | <b>0.197</b>      | 0.204              | 0.206           | 0.010               | <b>0.005</b>       | 0.007        | 605.9    | 52.6               | <b>19.2</b> |
| protein       | Pub +2.7%              | <b>1.722</b>      | 1.769              | 1.796           | <b>0.005</b>        | 0.005              | 0.007        | 1254.7   | 127.0              | <b>64.9</b> |

FM is the fastest Treeffuser sampler on 9/10 datasets and is especially valuable at scale: on protein the speedup is  $19.3\times$  over PUBLISHED and  $2.0\times$  over SCORE<sup>+</sup>; on california\_housing the corresponding speedups are  $31.6\times$  and  $2.7\times$ . On |cE|@90, however, SCORE<sup>+</sup> leads on 6/10 datasets and FM on 3/10, which is why Table 1 separates normalized-CRPS and calibration conclusions. Under this evaluation protocol, the per-dataset CRPS ranking reproduces the headline pattern: FM has the best average normalized CRPS, score<sup>+</sup> is the more reliable calibration row, and the published SDE remains best on wine, california housing, and protein.

## J Conditional Tail Calibration Diagnostic

The large-dataset inversion in Table 2 admits two plausible mechanisms: the fixed residualizer may be suboptimal at high  $n$ , and stochastic samplers may improve conditional tail calibration in high-

443 uncertainty regions. Table 15 reports direct 95% coverage on `california_housing` and `protein`,  
 444 including the highest predicted-IQR bin.

Table 15: High-uncertainty tail diagnostic on the two largest datasets. Cov@95 is marginal empirical coverage; Top-IQR Cov@95 is empirical coverage in the highest predicted-IQR quintile; both target 0.95. IQR-MACE is mean  $|cE|$  @95 across predicted-IQR quintiles, so lower is better. SDE-linear uses a linear stochasticity schedule; SDE- $\sqrt{t}$  uses the square-root schedule. All rows use the full training and test splits in a separate three-seed diagnostic.

| Dataset       | Variant             | CRPS         | Cov@95       | IQR-MACE     | Top-IQR Cov@95 |
|---------------|---------------------|--------------|--------------|--------------|----------------|
| cali. housing | Published SDE       | <b>0.207</b> | 0.958        | <b>0.015</b> | 0.931          |
|               | FM ODE-C            | 0.214        | <b>0.947</b> | 0.021        | 0.900          |
|               | FM ODE-E            | 0.213        | 0.947        | 0.021        | 0.900          |
|               | FM SDE-C linear     | 0.215        | 0.969        | 0.023        | 0.939          |
|               | FM SDE-C $\sqrt{t}$ | 0.215        | 0.967        | 0.022        | 0.938          |
|               | FM SDE-E linear     | 0.213        | 0.971        | 0.023        | <b>0.944</b>   |
| protein       | FM SDE-E $\sqrt{t}$ | 0.213        | 0.969        | 0.023        | <b>0.944</b>   |
|               | Published SDE       | 1.787        | 0.944        | 0.018        | 0.925          |
|               | FM ODE-C            | 1.860        | 0.933        | 0.025        | 0.923          |
|               | FM ODE-E            | <b>1.771</b> | <b>0.949</b> | <b>0.008</b> | 0.943          |
|               | FM SDE-C linear     | 1.859        | 0.959        | 0.011        | 0.960          |
|               | FM SDE-C $\sqrt{t}$ | 1.856        | 0.956        | 0.012        | <b>0.954</b>   |
|               | FM SDE-E linear     | 1.779        | 0.974        | 0.024        | 0.973          |
|               | FM SDE-E $\sqrt{t}$ | 1.776        | 0.971        | 0.021        | 0.971          |

445 The diagnostic supports a tradeoff rather than a one-sided conclusion. On `california_housing`,  
 446 the deterministic FM rows under-cover the high-IQR bin (0.900), while adding FM stochasticity  
 447 lifts this coverage to 0.938–0.944; the price is marginal over-coverage (0.967–0.971) and no gain  
 448 over the published SDE on CRPS or IQR-MACE. On `protein`, residualizer-E with the deterministic  
 449 ODE is already close to the nominal target and remains best on CRPS and IQR-MACE, whereas  
 450 the residualizer-E SDE rows over-cover both marginally and in the high-IQR bin. Residualizer-C  
 451 stochasticity is directionally useful for coverage, especially with the square-root schedule, but gives  
 452 up CRPS. Thus the safest interpretation is not that SDE samplers are dominated, but that their benefit  
 453 is conditional and must be selected against conditional-calibration metrics rather than headline CRPS  
 454 alone.

## 455 K Reproducibility

456 The submission includes an anonymized frozen repository via `anonymous.4open.science`. The  
 457 repository contains the implementation in `src/`, the benchmark harness in `benchmarks/`, the pa-  
 458 per sources in `paper/`, and the `pixi.toml/pixi.lock` environment used for the reported runs.  
 459 All sweeps in this paper are driven by YAML configs under `benchmarks/configs/`. Each re-  
 460 sult file is a JSON Lines document recording the full hyperparameter set, random seeds, the git  
 461 commit hash of the model code, a hash of the Treeffuser source tree, and every reported met-  
 462 ric on a per-dataset/per-fold row. The headline tuning runs persist their selected configurations  
 463 under `benchmarks/results/tuning/best_params/` and their five-fold evaluation rows under  
 464 `benchmarks/results/tuning/eval/`. The configs and result artifacts that produced the tables  
 465 are listed alongside in Table 16; the paper PDF is rebuilt with the paper `pixi` task.

Table 16: Provenance: appendix tables and the config / result file each was produced from.

| Table        | Config                                                                     | Result file                                                                      |
|--------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 6            | <code>fm_off_vs_mean_sweep.yaml</code>                                     | <code>fm_off_vs_mean_sweep_*.jsonl</code>                                        |
| 3            | <code>analytic summary</code>                                              | <code>-</code>                                                                   |
| 7            | <code>residualizer_fm_sweep.yaml</code>                                    | <code>residualizer_fm_sweep_*.jsonl</code>                                       |
| 8            | <code>fm_mean_scale_sweep.yaml</code>                                      | <code>fm_mean_scale_sweep_*.jsonl</code>                                         |
| 9            | <code>fm_t_sampling_sweep.yaml</code>                                      | <code>fm_t_sampling_sweep_*.jsonl</code>                                         |
| 10           | <code>fm_loss_weighting_sweep.yaml</code>                                  | <code>fm_loss_weighting_sweep_*.jsonl</code>                                     |
| 11           | <code>sde_lower_stoch_sweep.yaml, stochasticity_schedule_sweep.yaml</code> | <code>sde_lower_stoch_sweep_*.jsonl, stochasticity_schedule_sweep_*.jsonl</code> |
| 12           | <code>ode_vs_sde_path_ablation.yaml</code>                                 | <code>ode_vs_sde_path_ablation_*.jsonl</code>                                    |
| 13, 2, 14, 1 | <code>tuning_manifest.yaml</code>                                          | <code>benchmarks/results/tuning/eval/*.jsonl</code>                              |
| 15           | <code>paper_real_data_v2.yaml, tail_calibration_full.yaml</code>           | <code>paper_real_data_v2_full.jsonl, tail_calibration_full.jsonl</code>          |
| 4            | <code>benchmarks/tuning/search_spaces.py</code>                            | <code>-</code>                                                                   |

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